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SCHOOL OF SCIENCE AND ENGINEERING

CSC 4305 · Software Engineering · Fall 2026

Use Cases, Scenarios and Acceptance Criteria

Three forms and what each is for, the fully dressed use case, extensions derived by seven checks rather than by inspiration, and the line from an extension to a test

Session 11 of 28 · Tuesday 13 October 2026 · 1:00 – 2:20 PM

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SESSION 11 OF 28

Use Cases, Scenarios and Acceptance Criteria

Tuesday, 13 October 2026 · 1:00 – 2:20 PM · Unit 2 · Requirements

LEARNING OBJECTIVES

- ✓ Choose between a use case, a scenario and a user story for a given goal, and say what each one is bad at.
- ✓ Write a fully dressed use case — actors, stakeholders and interests, preconditions, guarantees, main success scenario, extensions.
- ✓ Derive alternate and exception flows with the seven-check enumeration, rather than by inspiration.
- ✓ Draw a use-case diagram with a correct boundary, and explain why the diagram carries less information than the text.
- ✓ Turn an extension into a Given/When/Then acceptance criterion that becomes a test.

FROM SESSIONS 9 AND 10

The SRS baseline exists: twelve functional requirements, eight NFRs with fit criteria, four domain requirements and a data dictionary. What none of it expresses is **behaviour over time** — what happens at step four when the power fails, and what the clerk sees when two children match. Today writes that down, in a form that becomes tests in Session 23.

Assemble SRS v0.1 — who writes which section

Applying Session 10. Every ingredient now exists somewhere; what is missing is one document and six names against it.

WHAT HAPPENS

Pair A — §2

Overall description: the post, the power, the devices, the users. Written so that NFR-01, 02 and 08 are obviously necessary to a stranger

Pair B — §4 and §5

Eight NFRs with their fit criteria; four DRs with the article cited

Pair C — appendices

Data dictionary (six entries now) and the traceability matrix: requirement → story → source → test id

1. **Agree the omissions first.** Pick the best “what this SRS does not contain” half-page from the six homework submissions, and adopt it as the front page. **Five minutes.**
2. **Each pair states its deadline** and what it needs from the other two. **Five minutes.**
3. **Count the TBDs** in the traceability matrix. That number is your test-writing backlog, and you will meet it again in Session 23. **Four minutes.**

Output: srs/SRS-v0.1 committed to the team repository, section owners and deadlines in the progression log, and the TBD count written down where you will find it later.

From now on: this is a **baseline**. A requirement that changes without a recorded reason is a defect.

1 · Three forms, three purposes

Use cases, scenarios and user stories are not three competing methods you pick between once. They answer different questions, and a mature team uses all three at the same time on the same system.

Form	What it is	Best for	What it is bad at
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Use case	A complete description of every way an actor can pursue one goal — including all the ways it fails	Completeness; discovering exception paths; anything contractual or regulated	Verbose; goes stale; tempts you into specifying the interface
Scenario	One concrete narrative, with real names and real data	Talking to stakeholders; making an abstraction believable; seeding a test	Covers exactly one path. It is evidence, not a specification
User story	A short placeholder for a conversation, plus acceptance criteria	Planning, prioritising, delivering incrementally	Exception paths vanish unless you deliberately go looking for them

WHY YOU NEED MORE THAN STORIES — STATED PLAINLY

A user story is small on purpose, and the price of that is the exceptions. S-04 — “register a patient” — is five points and three acceptance criteria. The *register a patient* use case has eleven extensions, and among them are a lost record, a duplicated patient and an under-18 registered without a guardian. Those three are where the legal and data-integrity risk of this entire system lives. Stories plan the sprint. Use cases are how you find out what the sprint forgot.

APPLIED TO SANTE-BASSAM

Which goal gets which form — you cannot afford a fully dressed use case for everything

Goal	Form	Why
Register a new patient (UC-03) · record a visit outcome (UC-04) · synchronise records (UC-08)	Fully dressed use case	The value of the system and every legal and data-integrity risk live in these three. Their exception paths are where a lost or duplicated health record comes from
Book or reschedule an appointment (UC-05) · produce the monthly report (UC-07)	Casual use case main scenario plus named extensions	Important, but the failures are recoverable and visible — somebody notices and does it again
“Madame Koné registers Awa, aged 2, from Modeste, whose mother has no card”	Scenario	Read aloud to the nurse at the review. It is also the seed for acceptance test AT-01
Print a card · edit the village list · change a phone number	Stories only	Small, low risk, no interesting exceptions. A use case here would be ceremony

2 · The fully dressed use case

Alistair Cockburn’s template. Each field earns its place, and the fields students delete as bureaucratic are reliably the ones doing the work — *stakeholders and interests* is where the conflicts surface, and *minimal guarantee* is where the power cut gets handled.

APPLIED TO SANTE-BASSAM

UC-03 Register a new patient — every field filled

UC-03 · REGISTER A NEW PATIENT

PRIMARY ACTOR	Clerk (also: CHW, on the tablet)		
SCOPE	SANTE-BASSAM patient registry	LEVEL	user goal
STAKEHOLDERS AND INTERESTS			
Clerk	finish in under 90 s; not be blamed later for a duplicate		
Patient/guardian	a record findable at every future visit, card or no card		
District officer	complete village and birth-year data for the report		
Nurse-in-charge	duplicates detected at entry, not silently created		
Ministry	identifier format and guardian consent respected (DR-01, D		
PRECONDITIONS			
Actor is authenticated (UC-09); device has >= 50 MB free			
MINIMAL GUARANTEE			
No partial record is persisted; the registry is unchanged			
SUCCESS GUARANTEE			
A Patient exists locally with a unique local id, marked pending if offline, and is findable by FR-02 immediately			
TRIGGER			
A person presents at the desk who is not in the register			
MAIN SUCCESS SCENARIO			
1. Clerk selects "register new patient".			
2. System presents the registration form, fields in card order (NFR-04).			
3. Clerk enters given name, family name, birth date, sex and village.			
4. System checks the entry against existing patients (FR-03).			
5. System reports no likely duplicate.			
6. Clerk optionally enters mother's name and phone.			
7. Clerk confirms.			
8. System commits the record locally before displaying anything (FR-04), assigns a local id, and shows the id with a confirmation.			
9. Clerk writes the id on the paper card and gives it to the patient.			
EXTENSIONS			
see section 3 -- derived, not invented			
FREQUENCY			
~12 per clinic day; ~30 on vaccination Fridays			
OPEN ISSUE			
who may correct a name after registration, and is that an audit event? -- nurse to decide, wk 8			

Read step 8 again. “Commits locally *before* displaying anything” is NFR-02 appearing inside a sequence of steps — which is precisely what the SRS could not say and the use case can.

3 · Deriving extensions systematically

Students write the main success scenario easily, then invent one or two extensions by inspiration and stop. Inspiration finds the failures you already imagined; the interesting ones are the others. So do it mechanically instead: take **each step in turn** and ask the same **seven questions** of it.

THE SEVEN CHECKS — APPLIED TO EVERY STEP, IN ORDER

- 1 **Wrong input.** The actor supplies something invalid, impossible or empty at this step.
- 2 **Abandonment.** The actor walks away, is interrupted, or the patient leaves, mid-step.
- 3 **The system cannot do its part.** Power, disk, network, battery, printer — the resource this step needs is gone.
- 4 **Nothing there.** The data this step expects does not exist — no match, empty list, missing village.
- 5 **Too much there.** The data exists more than once — two matches, an ambiguous name, a duplicate.
- 6 **Timing.** Somebody else changed it while this step was in progress, or the step took so long it timed out.
- 7 **Authority.** The actor is not permitted to do this, or a required consent or signature is absent.

APPLIED TO SANTE-BASSAM

The seven checks run across UC-03 — eleven extensions, none of them guessed

Ext.	Step · check	Condition	Handling
3a	3 · wrong input	Birth date unknown — the guardian knows only the year	Accept year only; store as 1 July; flag as approximate on the record
3b	3 · nothing there	The village is not in the list of 23	Offer the nearest match; if genuinely new, record and queue for the nurse to approve (FR-11)
3c	3 · authority	Patient is under 18 and no guardian is present	Registration is permitted; guardian field required before the first Visit (DR-02)
4a	4 · too much there	Two existing patients match closely	Show both with village and birth year; the clerk selects or declares a new patient; the decision is logged
5a	5 · too much there	A likely duplicate is found and the clerk overrides the warning	Permitted, with the override recorded and the clerk's id attached (FR-03, NFR-05)
7a	7 · abandonment	The patient leaves before the clerk confirms	Nothing is persisted — the minimal guarantee. The form is recoverable for 10 minutes
8a	8 · system cannot	Power fails between commit and confirmation	The record is committed; on restart the clerk sees it already exists. This is the extension NFR-02 was written for
8b	8 · system cannot	Local storage is full	Refuse before the form is shown, not after — hence the precondition
8c	8 · timing	Two devices generate the same local id offline	Impossible by construction — device prefix in the id (S-02). Recorded here so that nobody removes it later
8d	8 · timing	The CHW registered the same child this morning; both sync tonight	Both are retained; the duplicate is flagged for the nurse after sync, never auto-merged

9a

9 · system
cannot

No blank cards left at the desk

Out of scope for the software.

Recorded as an operational note in the
SRS §2

Extensions 8c and 8d were found by the timing check, and neither appeared in any interview. Extension 9a is the kind of thing the enumeration turns up that is not a software problem at all — write it down anyway, then say where it belongs.

4 · Use-case diagrams, and their limits

A use-case diagram shows actors outside a boundary, use cases inside it, and lines between them. That is nearly all it shows, and being clear about what it omits is the point of this section.

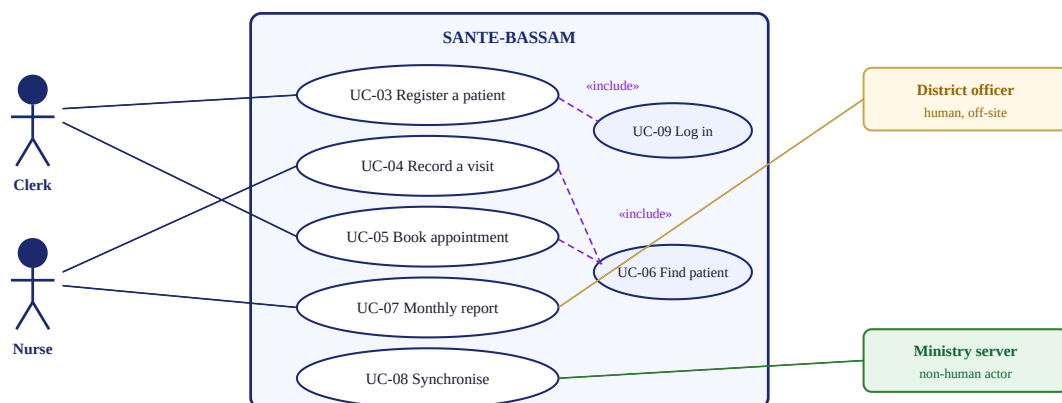


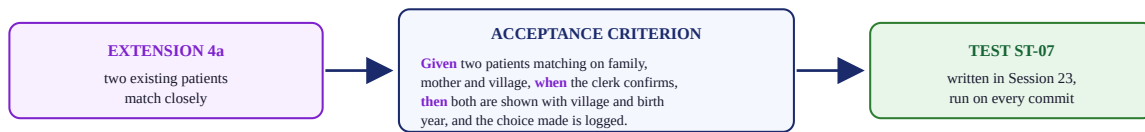
Figure 1. Two things students get wrong here. The boundary is the system, not the health post — the clerk stands outside it. And an actor need not be a person: the Ministry server is an actor because it participates in UC-08 and has interests of its own.

What the diagram cannot tell you

No sequence. No conditions. No data. No extensions — and the extensions are where every interesting requirement in §3 lives. The diagram is a **table of contents** for the text, useful on a wall in week 2 and useless as a specification in week 12. If your team produces the diagram and not the text, you have produced nothing.

5 • From extension to acceptance criterion

This is where Unit 2 closes the loop with Unit 1. An extension describes a condition and a handling; an acceptance criterion states the same thing in a form that can be executed. The translation is nearly mechanical.



Each arrow loses nothing. This is why the extensions were worth enumerating.

Figure 2. *Extension → criterion → test. Every extension in §3 that concerns software becomes at least one criterion; each criterion becomes at least one test. The eleven extensions of UC-03 are, between them, most of your acceptance test suite for registration.*

THE RULE THAT KEEPS THIS HONEST

An extension you cannot write as Given/When/Then is an extension you do not yet understand. Usually the missing piece is the **then** — you know something goes wrong but you have not decided what the system does about it. That undecided *then* is a question for the nurse, not a gap to fill in with a guess at your desk.

SESSION 11 TAKEAWAY

Stories plan the sprint; scenarios convince the stakeholder; use cases find what the stories forgot. Reserve the fully dressed form for the two or three goals where the value and the legal risk live, and for those goals derive the extensions mechanically — seven checks on every step — rather than by inspiration. Then translate each extension into Given/When/Then, and you have written a large part of your test suite twelve sessions before you write the tests.

6 • In-class exercise • run the seven checks

Twelve minutes in three pairs, four minutes comparing. Here is the main success scenario of UC-04, recording a visit outcome. It has no extensions yet.

UC-04 · Record a visit outcome — main success scenario

1. Nurse selects the patient from today's appointment list.
2. System shows the patient, their last visit and any open follow-up.
3. Nurse records the outcome: attended / missed / rescheduled.
4. Nurse optionally sets a follow-up date.
5. Nurse confirms.
6. System commits the Visit locally, then confirms on screen.
7. System makes the outcome visible to the pharmacy the same day (FR-12).

1. **Take two checks each.** Pair A takes *wrong input* and *abandonment*; Pair B takes *system cannot* and *nothing there*; Pair C takes *too much there* and *timing*. Everyone does *authority*.
2. **Run your checks across all seven steps**, in order. Write each extension as `<step><letter>`, a condition and a handling.
3. **Pick your best one and write it as Given/When/Then.**
4. **Report.** Two minutes each. The rest of the team says which extension they would have missed.

TWO THE SEVEN CHECKS WILL FIND THAT NOBODY GUESSES

Timing at step 7: the pharmacy reads the outcome, and then the nurse corrects it. What does the pharmacy see, and does anyone tell them? **Authority at step 3:** the clerk, not the nurse, is at the desk on Friday afternoon — may she record an outcome, and if she may not, what happens to the patient in front of her?

7 · Homework · individual · due before Session 12 (Friday 16 October)

TEMPLATE

Download **CSC4305_HW11_Template.docx** from **CSC_4305_SWE-Assignments**, fill it in, **rename** it **CSC4305_HW11_FirstName_LastName.docx** and **commit** it to **/homework/S11/** in **your own private repository**.

A .docx file. Not a PDF, not a link.

Q1 Finish UC-04. Run *all seven* checks across all seven steps, not just your pair's two. Produce the full extension table: reference, step and check, condition, handling.

Expect somewhere between eight and fourteen. If you have three, you ran the checks on step 3 and stopped.

Q2 Three acceptance criteria. Convert three of your extensions — the three you think are most likely to be got wrong in code — into Given/When/Then.

If you cannot write the *then*, that is a finding. Write the question you would ask the nurse instead, and mark it OPEN.

Q3 Choose the form. For each of these four SANTE-BASSAM goals, say whether it deserves a fully dressed use case, a casual one, a scenario or just a story — one sentence of justification each.

(a) A CHW registers a family of five during outreach. (b) The nurse corrects a misspelled patient name. (c) The officer downloads last month's report. (d) The clerk logs in at the start of the day.

Q4 Draw a use-case diagram for the CHW only — their actors, their use cases, the boundary, and at least one «include». Hand-drawn and photographed is fine.

Then one sentence underneath: name something important about the CHW's work that your diagram cannot express.

Q5 One scenario, with real names. Write the Madame Koné scenario in your own words — one paragraph, concrete, no jargon, the kind of thing you could read aloud to the nurse at a review. Then say which use case it exercises and which extension it hits.

Q6 Reading. Sommerville, *Software Engineering* 10th ed., §5.3 (use cases and interaction models). Optional: Cockburn, *Writing Effective Use Cases*, chapters 1–3.

Where Unit 2 ends. You arrived with sixteen guesses and you leave with a baselined specification, a dictionary, a use-case set and the beginnings of a test suite — all of it traceable to something somebody said, did or wrote down. Unit 3 turns it into structure: what the system is made of, and why it is made of those parts rather than others.

Key terms

use case

scenario

fully dressed

casual use case

primary actor

stakeholders and interests

precondition

minimal guarantee

success guarantee

main success scenario

extension

the seven checks

system boundary

non-human actor

«include» · «extend»

acceptance criterion

Readings and references

- Sommerville, I. *Software Engineering*, 10th ed. — Chapter 5, §5.3.
- Cockburn, A. *Writing Effective Use Cases*, 2000 — the fully dressed template and the extension enumeration.
- Fowler, M. *UML Distilled*, 3rd ed. — use-case diagrams and their limits.

- Adzic, G. *Specification by Example*, 2011 — scenarios as executable tests.